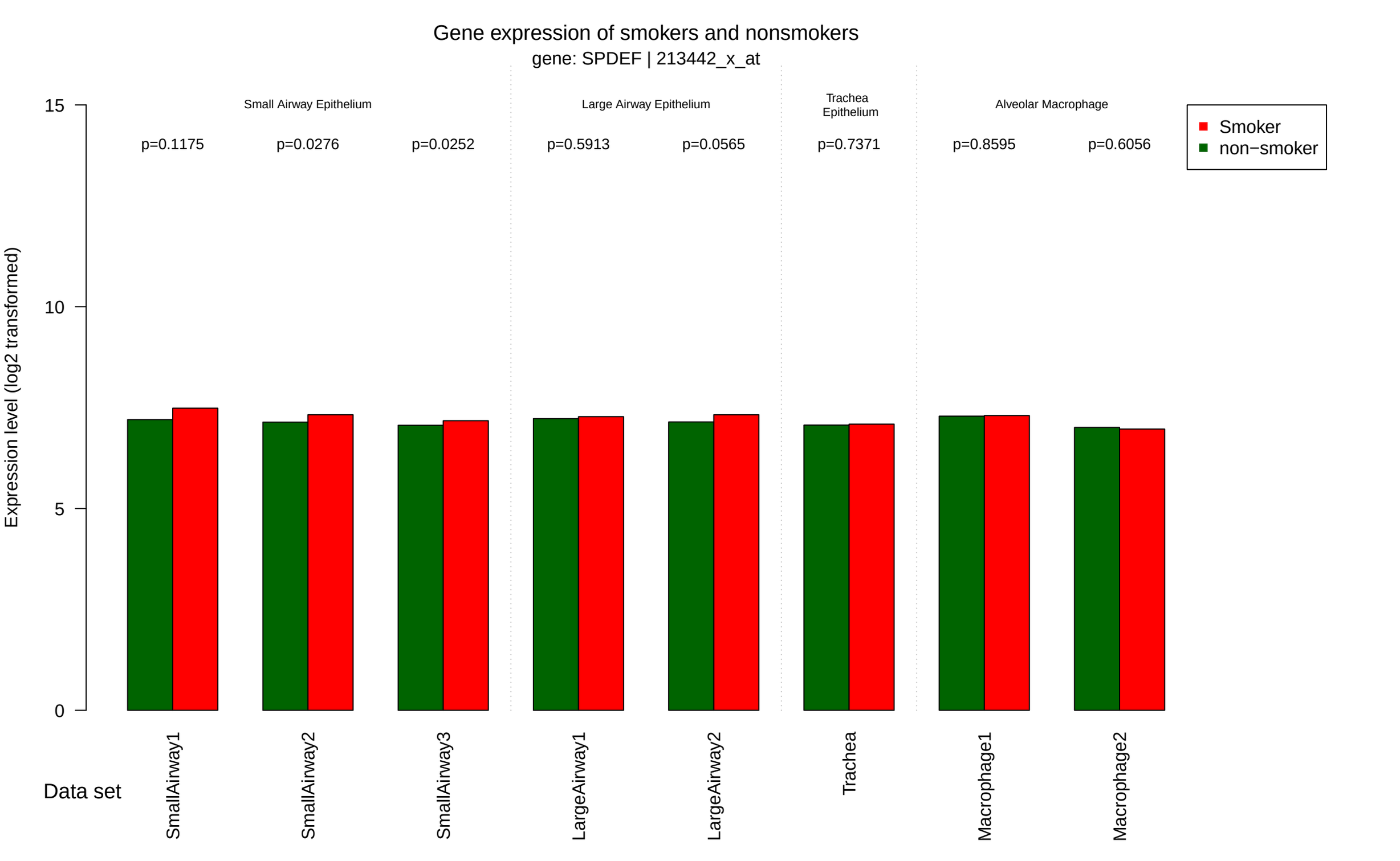
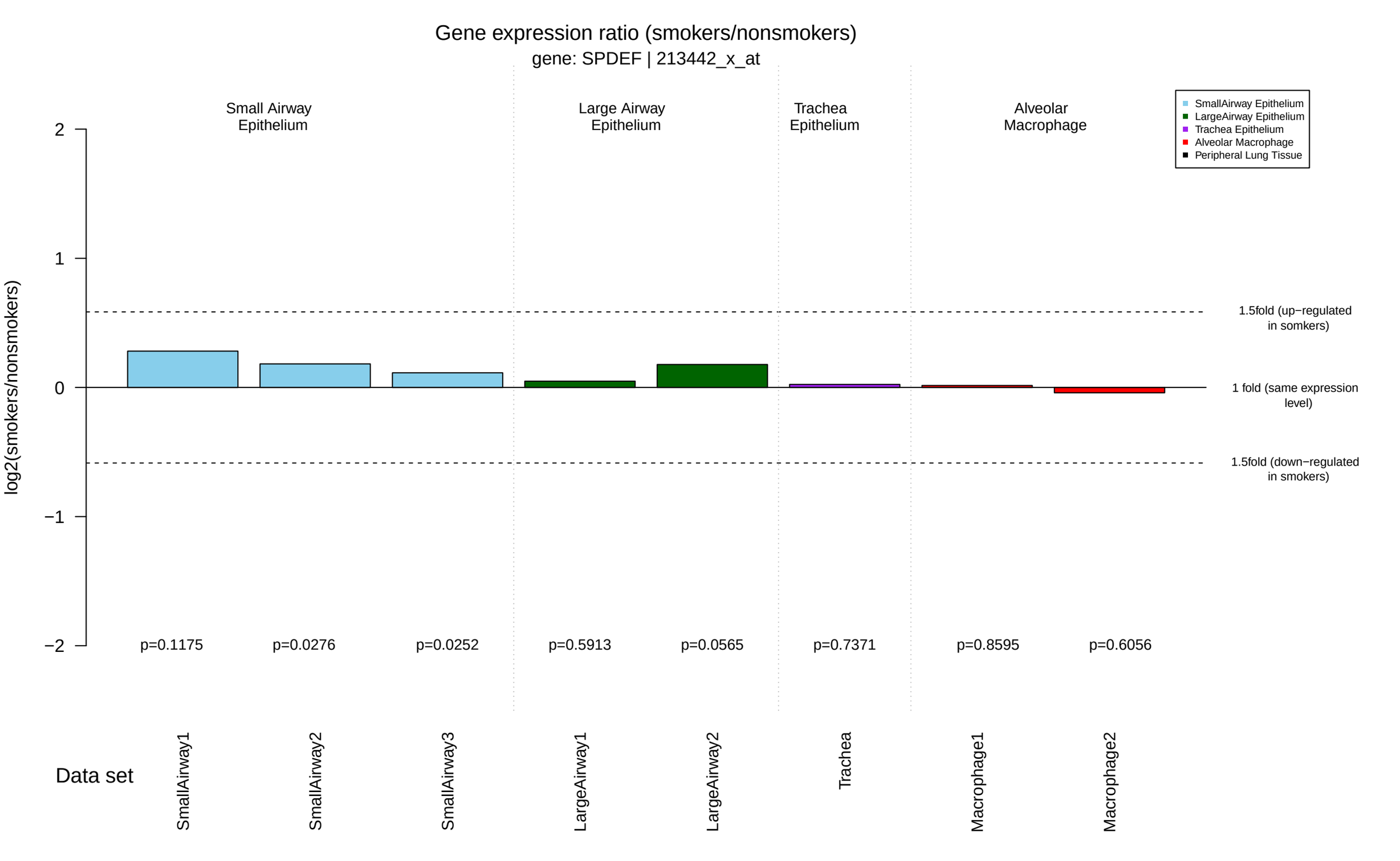
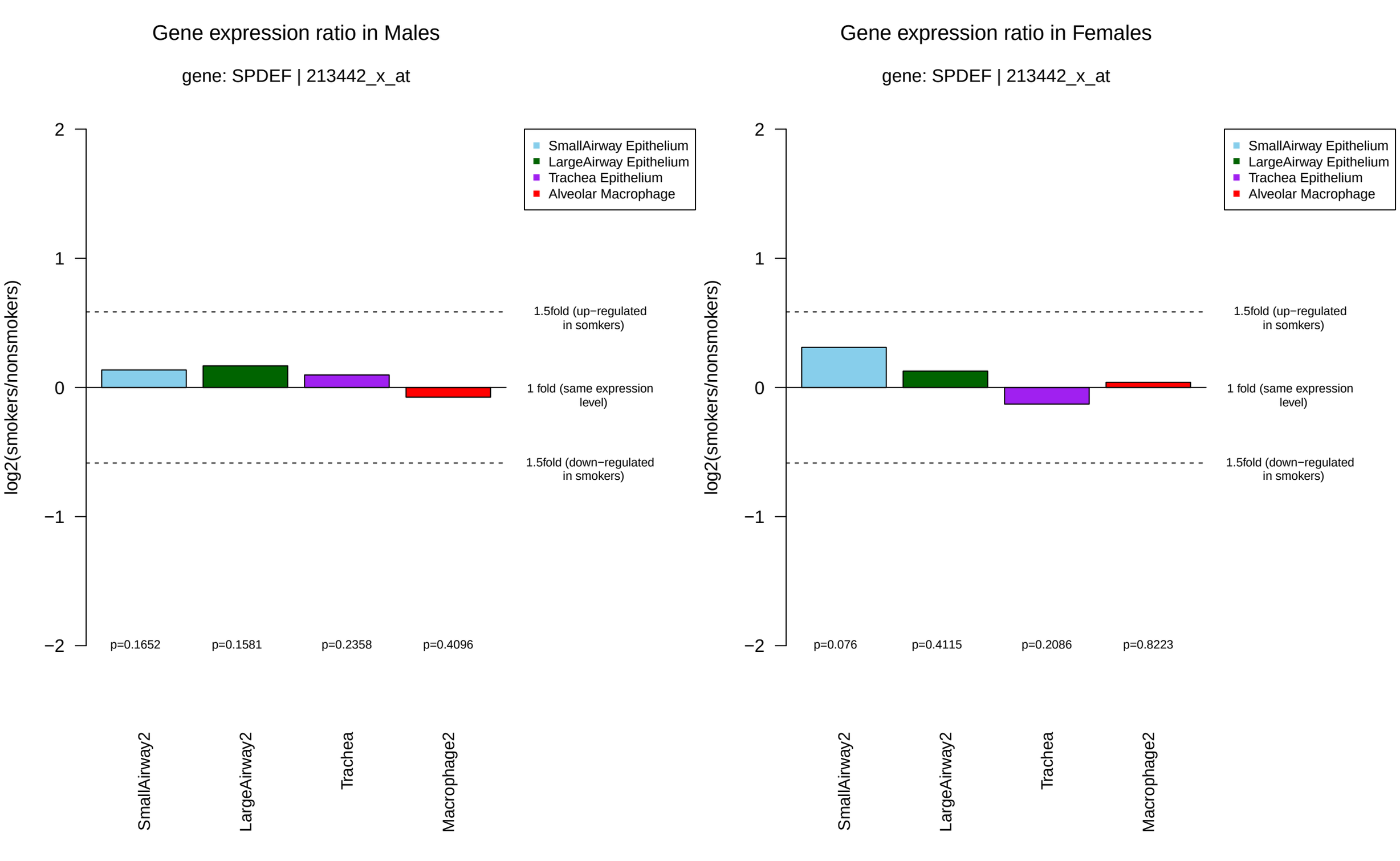
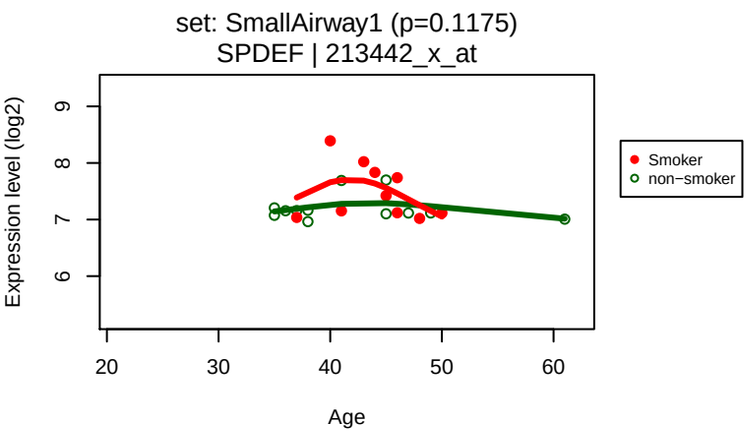
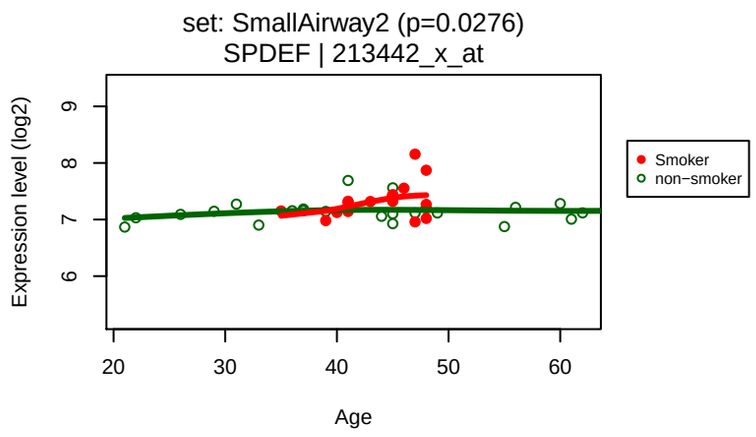
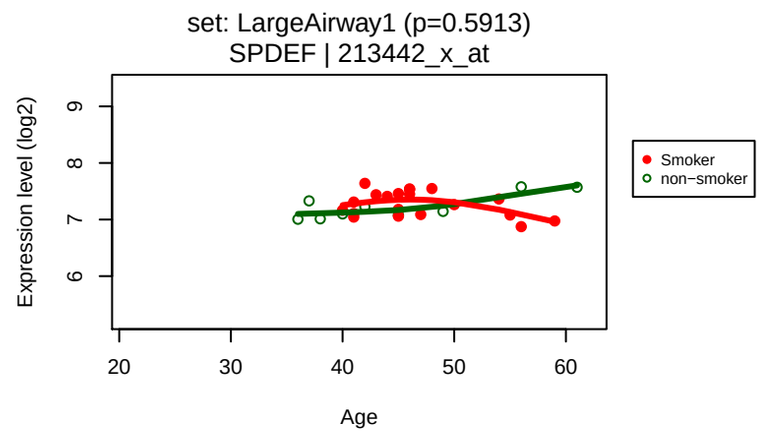

Gene expression of smokers and nonsmokers
gene: SPDEF | 213442_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1175
p=0.0276
p=0.0252
p=0.5913
p=0.0565
p=0.7371
p=0.8595
p=0.6056
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

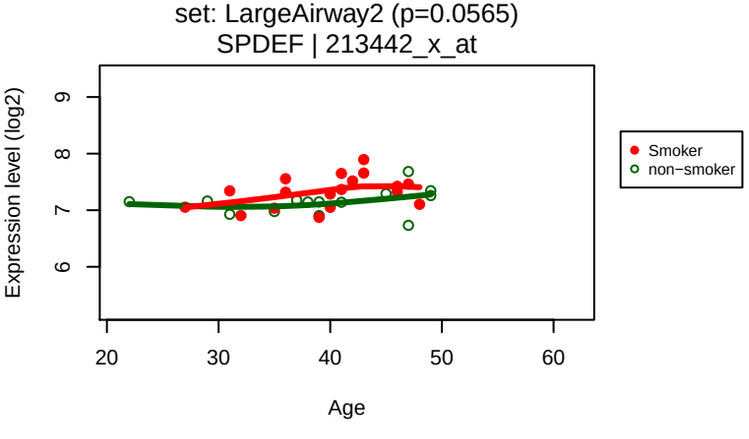

Gene expression ratio (smokers/nonsmokers)
gene: SPDEF | 213442_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1175
p=0.0276
p=0.0252
p=0.5913
p=0.0565
p=0.7371
p=0.8595
p=0.6056
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

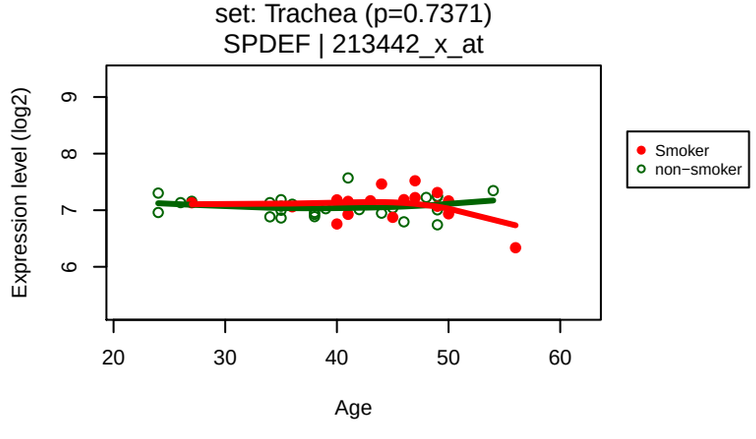

Gene expression ratio in Males
gene: SPDEF | 213442_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1652
p=0.1581
p=0.2358
p=0.4096
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPDEF | 213442_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.076
p=0.4115
p=0.2086
p=0.8223
SmallAirway2
LargeAirway2
Trachea
Macrophage2

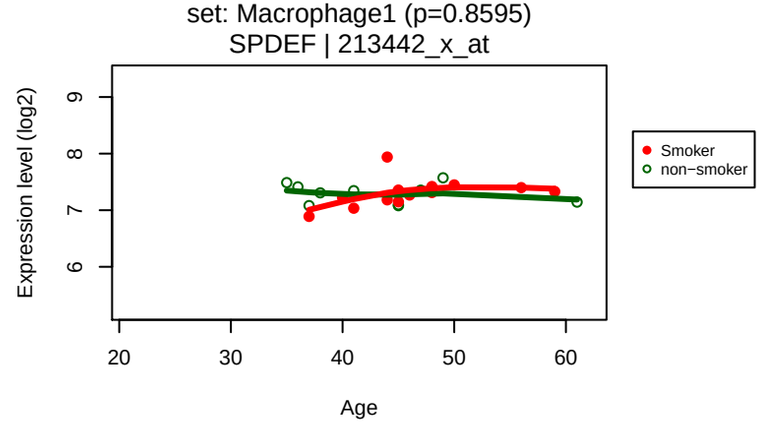

set: SmallAirway1 (p=0.1175)
SPDEF | 213442_x_at
Expression level (log2)
Age
Smoker
non−smoker

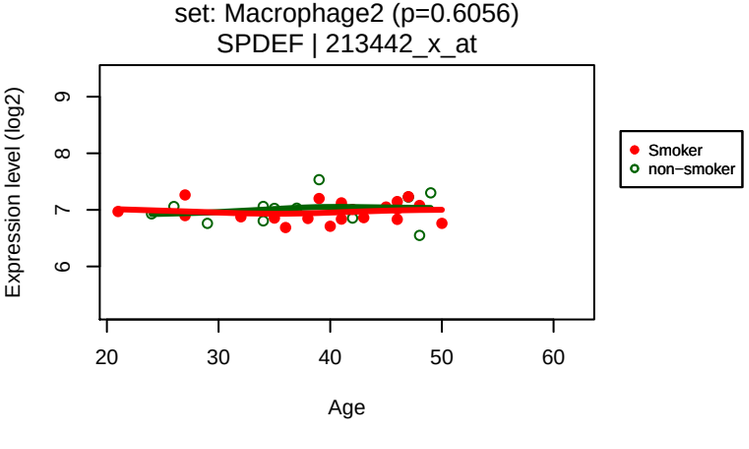

set: SmallAirway2 (p=0.0276)
SPDEF | 213442_x_at
Expression level (log2)
Age
Smoker
non−smoker

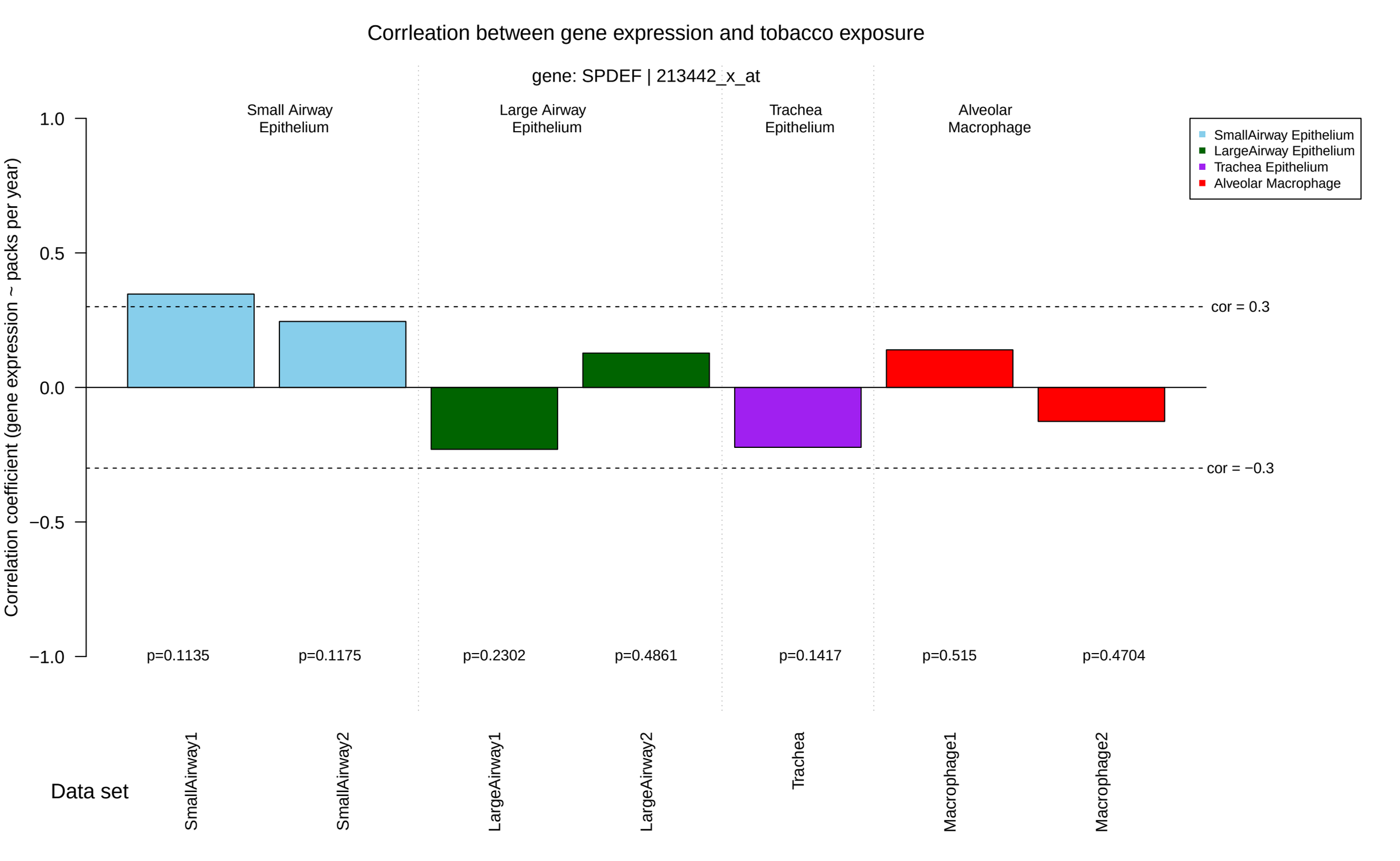

set: LargeAirway1 (p=0.5913)
SPDEF | 213442_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0565)
SPDEF | 213442_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7371)
SPDEF | 213442_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8595)
SPDEF | 213442_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6056)
SPDEF | 213442_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPDEF | 213442_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1135
p=0.1175
p=0.2302
p=0.4861
p=0.1417
p=0.515
p=0.4704
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2